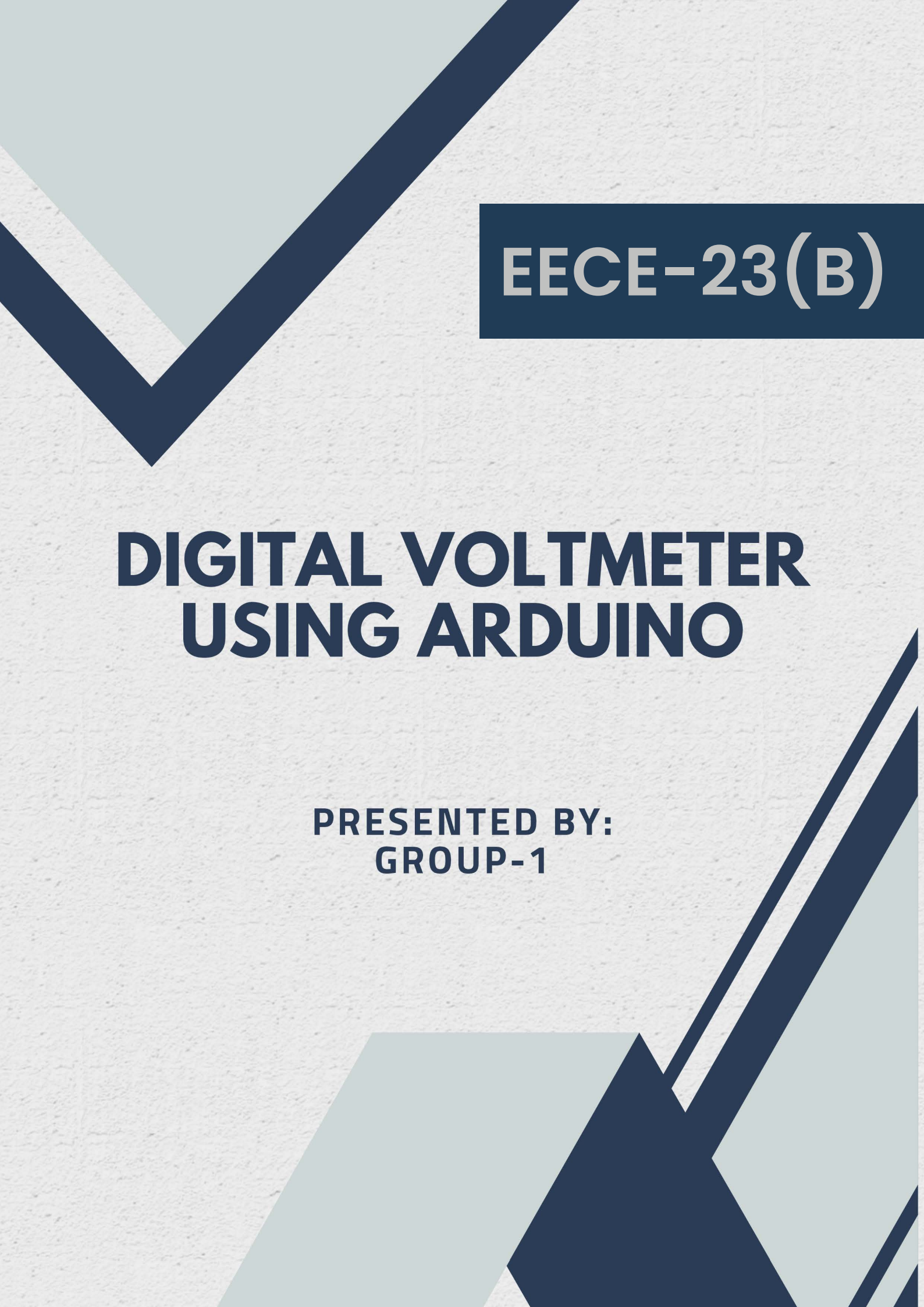




EECE-23(B)

DIGITAL VOLTMETER USING ARDUINO

**PRESENTED BY:
GROUP-1**

Digital Voltmeter Using Arduino and LCD Display

Contribution Table

Name	Roll	Contribution
Abdullah Al Kafi	202516060	Hardware Assembling
Mst. Sanjana Afrin Tanha	202516067	Report Preparation
Muhammad Labib Daiyan Bhuiyan	202516073	Code Compiling
Hasin Ishraq Khan	202516071	Presentation Preparation
Md. Rowshon Jamil Anondo	202516077	Equipment Purchasing

Objective

The primary objectives of this project are:

- To understand the concept of **analog-to-digital conversion (ADC)** and its implementation using Arduino.
- To design and implement a **voltage divider circuit** for safe measurement of higher voltages.
- To establish proper **interfacing between Arduino and an LCD display** for real-time output.
- To ensure safe handling of electrical components while measuring voltages.

Introduction

A voltmeter is a device used to measure the potential difference between two points in an electrical circuit. Traditional voltmeters operate on analog principles, but digital voltmeters (DVMs) have gained prominence due to their higher accuracy, readability, and ease of integration into electronic systems.

This project focuses on building a **digital voltmeter using an Arduino Uno R3 and a 16×2 LCD display**. The system employs a **voltage divider circuit** to step down input voltages to a range suitable for the Arduino's ADC (0–5 V). The Arduino then converts this analog signal into a digital value, processes it, and displays the corresponding voltage on the LCD screen.

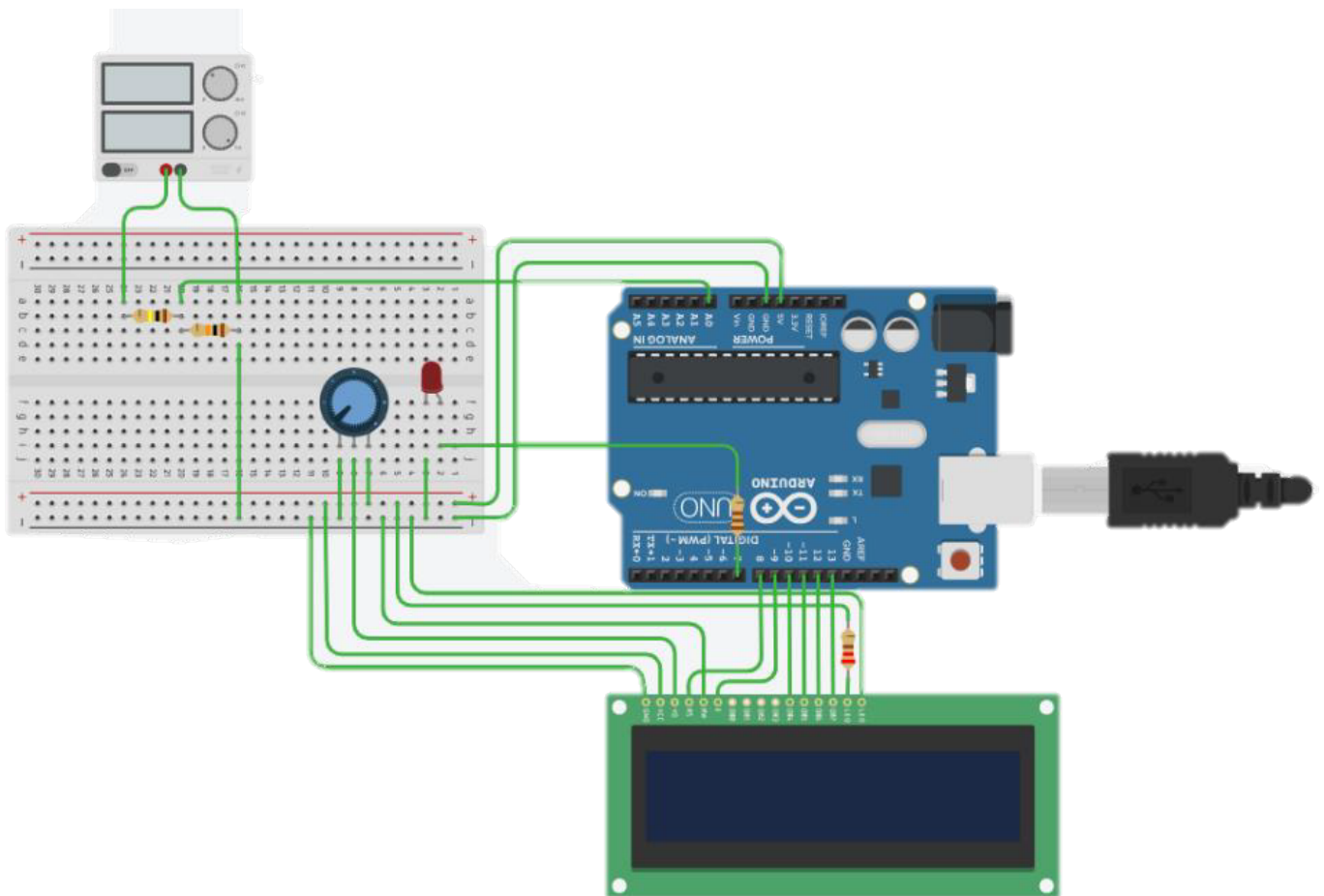
Such a system demonstrates practical applications of microcontrollers in measurement systems, offering accuracy, cost-effectiveness, and ease of use compared to conventional instruments.

Components

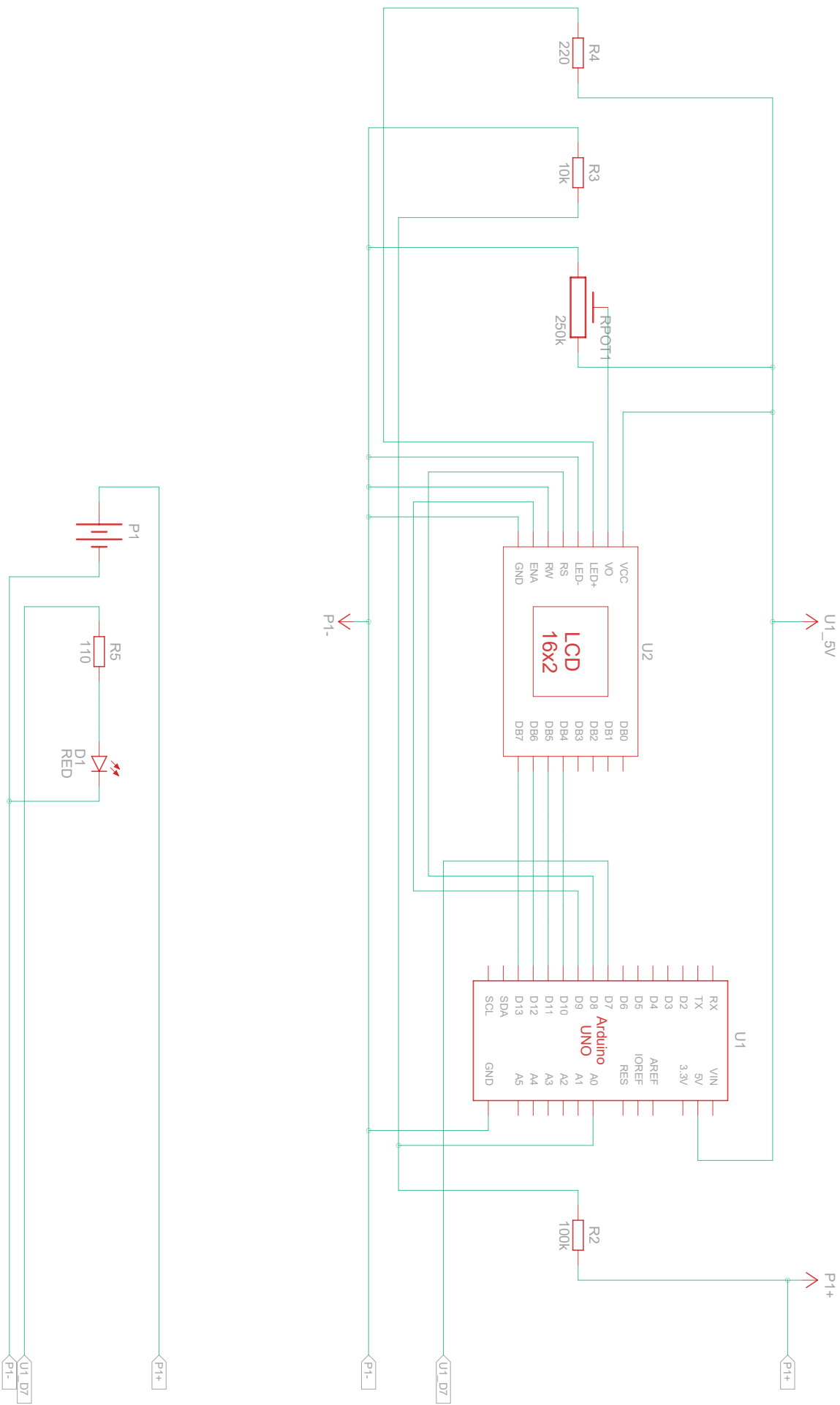
The components required for this project are:

1. **Arduino Uno R3** – Microcontroller board for processing input digital and analog signals.
2. **Jumper wires (male–female, male–male)** – For circuit connections.
3. **Resistors (220 Ω , 10k Ω , 100k Ω)** – For voltage divider circuit and limiting current through LCD.
4. **Potentiometer (10k Ω)** – To adjust LCD contrast.
5. **LED bulb** – For indication/testing purposes.
6. **16 $\times$ 2 LCD display** – For displaying voltage readings.
7. **Breadboard** – For circuit assembly and prototyping.

Circuit Diagram



Block Diagram



Methodology

1. Voltage Scaling:

Since the Arduino Uno can measure only up to 5 V, a voltage divider circuit was designed using two resistors (R_1 and R_2). This ensured that higher input voltages (e.g., 12 V, 24 V) were safely reduced to a range suitable for the Arduino's ADC pin. The ratio of R_1 and R_2 was chosen carefully to cover the required voltage range while protecting the microcontroller.

Formula:

$$V_{in} = V_{out} \times \frac{R_1 + R_2}{R_2}$$

2. Analog-to-Digital Conversion (ADC):

The Arduino Uno is equipped with a 10-bit ADC that converts voltages between 0–5 V into digital values between 0 and 1023. This means each step corresponds to approximately 4.9 mV. For instance, if the ADC gives a value of 512, it represents about 2.5 V at the input pin.

3. Voltage Calculation in Code:

The Arduino sketch (code) reads the ADC value using the `analogRead()` function. Using the known resistor ratio from the voltage divider, the program scales this value back to the actual input voltage. The formula implemented in the code is:

$$V_{in} = \frac{ADC_{value} \times 5.0}{1023} \times \frac{R_1 + R_2}{R_2}$$

4. LCD Interfacing:

A 16×2 LCD display was connected to the Arduino using 4-bit mode for efficient pin usage. A 10kΩ potentiometer was used to adjust the contrast of the display. The processed voltage values were displayed in real time, updating whenever the input voltage changed.

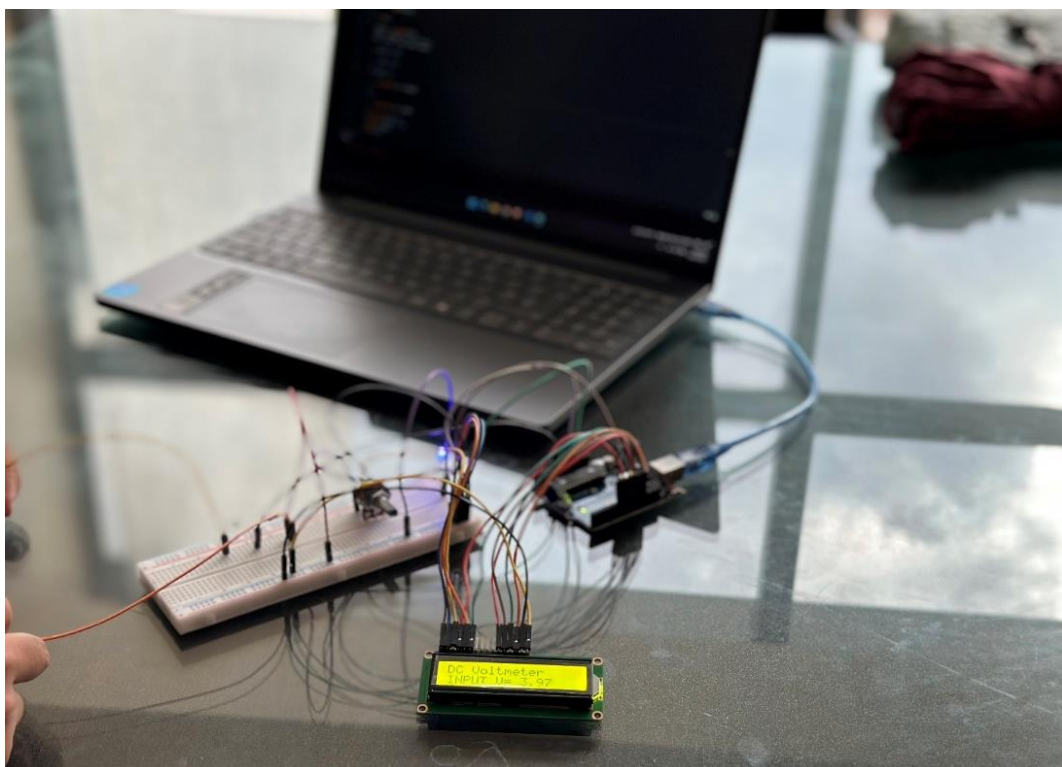
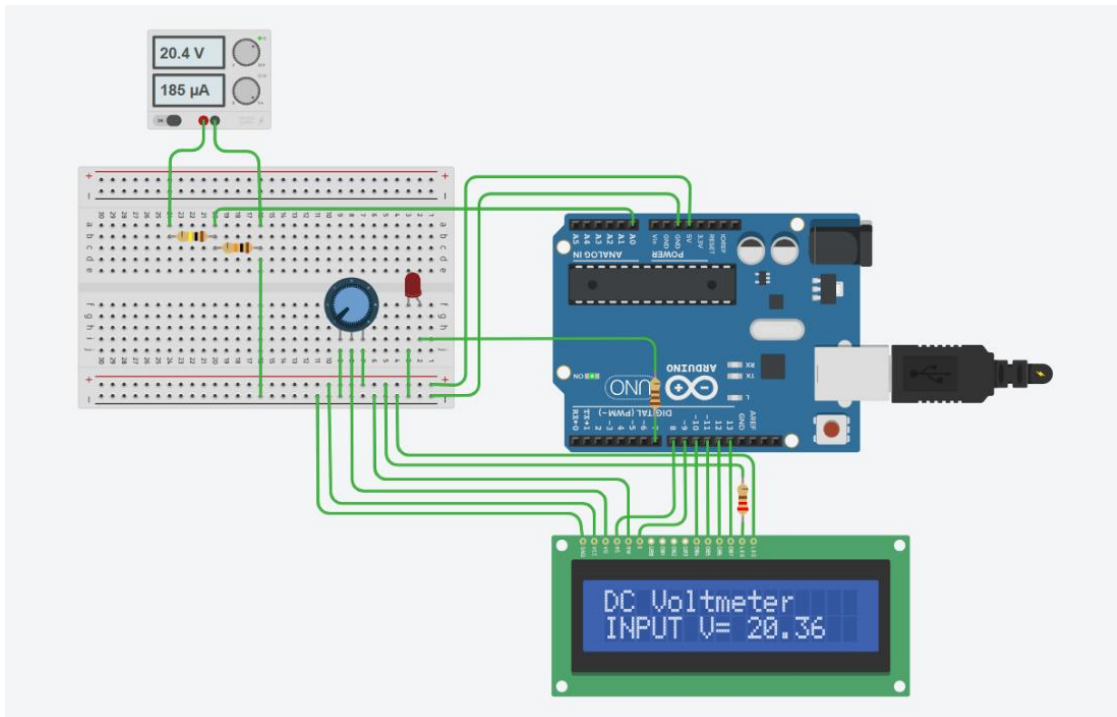
5. Testing and Validation:

The system was first tested with low voltages (1–5 V) and the readings were compared with a standard digital multimeter. Later, higher voltages (up to 35 V) were applied through the divider, and the output was verified. The readings were found to be consistent and accurate within a small error margin.

Simulation Result & Output

The simulation (using Proteus/TinkerCAD) confirmed that the system works correctly. The LCD updates automatically when the input voltage changes. For example:

- An input of 20.4 V produced a display reading of approximately 20.36 V.
- The system remained stable for voltages ranging from 0–35 V (with appropriate resistor values).



Result Analysis

The results obtained from the project can be summarized as follows:

- The Arduino successfully read the input voltage (after scaling) and displayed it correctly on the LCD in real time.
- The readings were stable with minimal fluctuations under steady input voltage conditions.
- For voltages within 0–5 V (directly measurable without divider), the readings matched a commercial digital multimeter with negligible error.
- For voltages above 5 V (measured using the voltage divider), the accuracy depended on the precision of the resistor values.
- The system provided an accuracy within ± 0.1 V for most test cases, which is acceptable for educational and general laboratory applications.
- The Arduino's 10-bit ADC provided a resolution of approximately 4.9 mV per step, which was sufficient for DC voltage measurements.
- The LCD display provided clear and immediate updates, making the system user-friendly.
- Small deviations were observed due to resistor tolerance ($\pm 5\%$), which slightly affected higher voltage readings.
- The accuracy also varied slightly when the Arduino's supply voltage fluctuated, showing the importance of a stable 5 V power source.
- The system could not measure AC voltages directly without additional circuitry (rectifier and filter).
- The maximum measurable voltage was limited by the chosen resistor ratio — if improperly selected, it could risk damaging the Arduino.

Discussion

The digital voltmeter designed in this project successfully demonstrates the use of a microcontroller-based system for voltage measurement. The Arduino Uno, being an open-source and widely supported microcontroller, provided a flexible and cost-effective solution. By combining hardware (voltage divider circuit, LCD, resistors) with software (Arduino IDE for coding ADC and calculation), the system was able to replicate the functionality of a commercial voltmeter at a fraction of the cost (almost less than 20%).

One of the most important observations is the dependence of the system's accuracy on the resistor values used in the voltage divider. Resistors with $\pm 5\%$ tolerance may introduce noticeable errors, especially at higher voltages. This limitation can be overcome by using precision resistors ($\pm 1\%$ or better) or by calibrating the device in software.

Another important factor is the resolution of the Arduino's 10-bit ADC. Since it divides the 0–5 V range into 1024 discrete steps, the smallest measurable change is about 4.9 mV. While this is sufficient for most general-purpose applications, higher-resolution ADCs (12-bit, 16-bit, or more) could be used for applications requiring very high accuracy.

The project also highlights the limitations of using Arduino's default internal reference voltage. Variations in the 5 V supply can lead to inconsistent readings. An external voltage reference IC would help improve stability and accuracy.

Despite these limitations, the digital voltmeter achieved its primary goals: it displayed accurate readings, updated in real time, and handled voltages beyond the Arduino's 5 V limit through the voltage divider. This makes it a highly educational project that bridges theoretical knowledge (Ohm's law, voltage division rule, ADC principles) with practical implementation.

Conclusion

The project has been successfully designed, implemented, and tested. It fulfills the intended objectives of demonstrating analog-to-digital conversion, voltage scaling through a divider, and real-time display of results.

Through this project, the following outcomes were achieved:

- A functional digital voltmeter capable of measuring DC voltages beyond the Arduino's direct input range of 5 V.
- An understanding of how voltage divider circuits safely adapt higher voltages for microcontrollers.
- Practical experience with Arduino coding, ADC utilization, and LCD interfacing.
- Validation of the system through simulations and hardware testing, which confirmed the accuracy and reliability of results.

The project demonstrates that low-cost microcontrollers like Arduino can be effectively used in measurement and instrumentation applications. While not a replacement for industrial-grade equipment, the prototype provides valuable insights into circuit design, programming, and system integration.

Overall, this project contributes both to learning and to practical application. It can serve as the foundation for more advanced systems, such as multi-functional measurement devices, IoT-enabled monitoring systems, or portable lab instruments.

Future Scope

- IoT Integration: Sending voltage data to cloud platforms for remote monitoring.
- Data Logging: Storing readings in an SD card for analysis.
- AC Measurement: Expanding the system to measure AC voltages using rectifiers.
- Higher Accuracy: Using precision resistors and external ADC modules.
- Compact Design: Making a handheld portable digital voltmeter device.